https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	t is the independent variable and T_C is the dependent variable **or** vary t and measure T_C	**1**
	keep T_R constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • solid cylinder cooling • insulation surrounding all of the cylinder • thermometer touching cylinder inside insulation • insulation and thermometer labelled	**1**
	method to heat the cylinder uniformly, e.g. place in oven/immerse in hot water **or** diagram showing cylinder in oven or hot water	**1**
	method to determine time t, e.g. stopwatch or temperature sensor connected to a data logger	**1**
	method to measure L e.g. use a ruler/calipers/micrometer **and** method to measure d e.g. use calipers/micrometer	**1**
	Method of Analysis	
	plot a graph of ln ($T_C - T_R$) against t or equivalent	**1**
	$U = -\frac{mc \times \text{gradient}}{A}$	**1**
	$Z = e^{y\text{-intercept}}$	**1**

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Question	Answer		Marks
1	**Additional detail including safety considerations**		**6**
	D1	precaution to prevent burns or use of hot cylinder / oven / hot water e.g. use of gloves, use of tongs	
	D2	keep thickness of the insulating material constant (for each T_C)	
	D3	method to measure *m*, e.g. use a (top-pan) balance	
	D4	for water bath/oven methods, wait for initial temperature of the cylinder to become uniform or constant throughout the cylinder	
	D5	(surface) $A = \pi dL + \frac{\pi d^2}{2}$ **or** $\pi dL + 2\left(\frac{\pi d^2}{4}\right)$	
	D6	repeat measurements of *d* along the length of the cylinder / in different directions **and** determine the average value of *d*	
	D7	description of how *c* is determined from a separate experiment by heating the cylinder using electrical heater **and** $c = \frac{\Delta E}{m\Delta\theta}$	
	D8	method of determining energy supplied to electrical heater to determine *c*, e.g. use of joulemeter for ΔE or electrical method using ammeter and voltmeter to determine IVt	
	D9	use several temperature sensors **and** determine the average T_C	
	D10	relationship valid if a straight line is produced (with *y*-intercept = ln *Z*) Do not accept line passing through the origin.	

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